

A Combined Levy–Gaussian Fourier Multiplier Proof via General Weak Differential Subordination

Lane 19 packet for arXiv:1706.01731

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Status

This packet records a likely valid full solution to the concrete multiplier-combination mechanism suggested on page 3 of Yaroslavtsev’s paper [1]. The result gives a direct proof, using the general weak differential subordination theorem from that paper, of the mixed jump/continuous Fourier multiplier class stated in [2, Theorem 4.1]. The obtained constant is the general weak-differential-subordination constant

$$C_{p,X} := \beta_{p,X}^2(\beta_{p,X} + 1),$$

not the sharper constant $\beta_{p,X}$ available from the special proof in [2].

This is not a claim to solve every open direction in [1]. The sharp continuous weak-subordination constant problem and the broad stochastic-integral BDG problem for arbitrary martingale noise remain separate.

Source Question

The source paper says:

Alternative approaches to Fourier multipliers for functions with values in UMD spaces have been constructed from the differential subordination for purely discontinuous martingales [...] and for continuous martingales [...]. It remains open whether one can combine these two approaches using the general weak differential subordination theory.

that sharp L^p -estimates for weakly differentially subordinated purely discontinuous martingales imply sharp estimates for the norms of a broad class of Fourier multipliers on $L^p(\mathbb{R}^d; X)$. Also there is a strong connection between the weak differential subordination of continuous martingales and the norm of the Hilbert transform on $L^p(\mathbb{R}; X)$ (see [41] and Remark 4.6).

Alternative approaches to Fourier multipliers for functions with values in UMD spaces have been constructed from the differential subordination for purely discontinuous martingales (see Bañuelos and Bogdan [4], Bañuelos, Bogdan and Bielszowski [3], and recent work [41]), and for continuous martingales (see McConnell [26] and Geiss, Montgomery-Smith and Saksman [18]). It remains open whether one can combine these two approaches using the general weak differential subordination theory.

2. PRELIMINARIES

In the sequel we will omit proofs of some statements marked with a star (e.g.

The Concrete Result

Let V be a Lévy measure on $\mathbb{R}^d$, let μ be a finite positive Borel measure on S^{d-1} , and let ϕ and ψ be scalar Borel functions with $|\phi| \leq 1$ and $|\psi| \leq 1$. Put

$$\Lambda(\xi) = \int_{\mathbb{R}^d} (1 - \cos \xi \cdot z) V(dz) + \frac{1}{2} \int_{S^{d-1}} (\xi \cdot \theta)^2 \mu(d\theta),$$

and define $m(\xi) = 0$ when $\Lambda(\xi) = 0$, while otherwise

$$m(\xi) = \frac{\int_{\mathbb{R}^d} (1 - \cos \xi \cdot z) \phi(z) V(dz) + \frac{1}{2} \int_{S^{d-1}} (\xi \cdot \theta)^2 \psi(\theta) \mu(d\theta)}{\Lambda(\xi)}.$$

This is the same mixed Lévy/Gaussian symbol class appearing in [2, Theorem 4.1]; the relevant page is shown below.

Here the answer will be given for m of quite a special form and X with the UMD property.

Let V be a Lévy measure on $\mathbb{R}^d$, that is $V(\{0\}) = 0$, $V \neq 0$ and

$$\int_{\mathbb{R}^d} (|x|^2 \wedge 1) V(dx) < \infty.$$

Let $\phi \in L^\infty(\mathbb{R}^d; \mathbb{C})$ be such that $\|\phi\|_{L^\infty(\mathbb{R}^d; \mathbb{C})} \leq 1$. Also let $\mu \geq 0$ be a finite Borel measure on the unit sphere $S^{d-1} \subset \mathbb{R}^d$, and $\psi \in L^\infty(S^{d-1}; \mathbb{C})$ satisfies $\|\psi\|_{L^\infty(S^{d-1}; \mathbb{C})} \leq 1$.

In the sequel we set $\frac{a}{0} = 0$ for each $a \in \mathbb{C}$. The following result extends [2, Theorem 1.1] to the UMD Banach space setting.

Theorem 4.1. *Let X be a UMD Banach space. Then the Fourier multiplier T_m with a symbol*

$$(4.2) \quad m(\xi) = \frac{\int_{\mathbb{R}^d} (1 - \cos \xi \cdot z) \phi(z) V(dz) + \frac{1}{2} \int_{S^{d-1}} (\xi \cdot \theta)^2 \psi(\theta) \mu(d\theta)}{\int_{\mathbb{R}^d} (1 - \cos \xi \cdot z) V(dz) + \frac{1}{2} \int_{S^{d-1}} (\xi \cdot \theta)^2 \mu(d\theta)}, \quad \xi \in \mathbb{R}^d,$$

has a bounded extension on $L^p(\mathbb{R}^d; X)$ for $1 < p < \infty$. Moreover, then for each $f \in L^p(\mathbb{R}^d; X)$

$$(4.3) \quad \|T_m f\|_p \leq \beta_{p,X} \|f\|_p.$$

Remark 4.2. The coefficient $\frac{1}{2}$ in both numerator and denominator of (4.1), even though it looks wired and useless (because one can always transform μ to 2μ), exists

Theorem 1 (combined WDS proof of the mixed multiplier bound). *Let X be a real or complex UMD Banach space and $1 < p < \infty$. Then the Fourier multiplier T_m , initially defined on $\mathcal{S}(\mathbb{R}^d) \otimes X$, extends to a bounded operator on $L^p(\mathbb{R}^d; X)$ and*

$$\|T_m f\|_{L^p(\mathbb{R}^d; X)} \leq \beta_{p,X}^2 (\beta_{p,X} + 1) \|f\|_{L^p(\mathbb{R}^d; X)}.$$

Proof Intuition

The pure-jump proof uses a compound Poisson process and transforms each jump increment by a scalar coefficient $\phi(z)$. The continuous proof uses a Brownian martingale and transforms each

directional Brownian integrand by a coefficient $\psi(\theta)$. Put both noises in the same Levy process. The untransformed parabolic martingale has a continuous quadratic-variation part and a jump quadratic-variation part. The transformed martingale multiplies the two corresponding pieces by contractions, so every scalar projection has smaller quadratic variation. Thus the transformed martingale is weakly differentially subordinate to the original one. The general theorem of [1] applies once to this mixed martingale, and the standard parabolic pairing computation gives exactly the quotient symbol above.

Proof

We give the proof first for a finite symmetric jump measure ν and finite measure μ , with bounded Borel coefficients. The standard truncation and symmetrization step at the end gives the stated Levy-measure formulation.

Let $L = (L_t)_{t \geq 0}$ be the $\mathbb{R}^d$ -valued Levy process whose characteristic exponent is

$$\mathbb{E} e^{i\xi \cdot L_t} = e^{-t\Lambda(\xi)}, \quad \Lambda(\xi) = \int_{\mathbb{R}^d} (1 - \cos \xi \cdot z) \nu(dz) + \frac{1}{2} \int_{S^{d-1}} (\xi \cdot \theta)^2 \mu(d\theta).$$

Equivalently, $L = L^J + L^G$, where L^J is a symmetric compound Poisson process with jump intensity ν , and

$$L_t^G = \int_{S^{d-1}} \theta W(d\theta, t)$$

for an $L^2(S^{d-1}, \mu)$ -cylindrical Brownian motion W . For $s < t \leq 0$ write $L_{s,t} = L_t - L_s$. If $f \in C_c^\infty(\mathbb{R}^d; X)$, set

$$P_{t,0}f(x) = \mathbb{E}f(x + L_{t,0}), \quad M_t = P_{t,0}f(x + L_{s,t}), \quad s \leq t \leq 0.$$

Then M is the usual parabolic X -valued martingale and $M_0 = f(x + L_{s,0})$.

Define a transformed martingale N by applying ϕ to jump increments and ψ to Brownian directional integrands:

$$\begin{aligned} N_t &= \int_s^t \int_{\mathbb{R}^d} (P_{r,0}f(x + L_{s,r-} + z) - P_{r,0}f(x + L_{s,r-})) \phi(z) \tilde{N}^J(dr, dz) \\ &\quad + \int_s^t \int_{S^{d-1}} \partial_\theta P_{r,0}f(x + L_{s,r}) \psi(\theta) W(d\theta, dr), \end{aligned}$$

where $\tilde{N}^J$ is the compensated jump measure of L^J and ∂_θ denotes the directional derivative. The corresponding untransformed martingale differences are obtained by replacing ϕ, ψ with 1; their sum is $M_t - M_s$.

Fix a real continuous linear functional $x^* \in X^*$. The quadratic variation of $\langle N, x^* \rangle$ on $[s, t]$ is the sum of

$$\int_s^t \int_{\mathbb{R}^d} |\langle P_{r,0}f(x + L_{s,r-} + z) - P_{r,0}f(x + L_{s,r-}), x^* \rangle|^2 |\phi(z)|^2 N^J(dr, dz)$$

and

$$\int_s^t \int_{S^{d-1}} |\langle \partial_\theta P_{r,0}f(x + L_{s,r}), x^* \rangle|^2 |\psi(\theta)|^2 \mu(d\theta) dr.$$

Since $|\phi| \leq 1$ and $|\psi| \leq 1$, this is bounded by the corresponding quadratic variation of $\langle M, x^* \rangle$. Also $N_s = 0$, while $|\langle N_s, x^* \rangle| \leq |\langle M_s, x^* \rangle|$. Hence N is weakly differentially subordinate to M . Yaroslavl'tsev's general weak differential subordination theorem [1, Theorem 4.1] therefore gives

$$(\mathbb{E}\|N_0\|^p)^{1/p} \leq C_{p,X}(\mathbb{E}\|M_0\|^p)^{1/p}.$$

After integration in x and translation invariance,

$$\int_{\mathbb{R}^d} \mathbb{E}\|N_0(x; s, 0; f)\|^p dx \leq C_{p,X}^p \|f\|_{L^p(\mathbb{R}^d; X)}^p.$$

Pair $N_0(x; s, 0; f)$ against a test function $g(x + L_{s,0})$, integrate in x , and use the Ito isometry/compensation identities for the jump and Brownian parts. The same calculation as in the Bañuelos–Bogdan parabolic proof gives

$$\int_{\mathbb{R}^d} \mathbb{E}\langle N_0(x; s, 0; f), g(x + L_{s,0}) \rangle dx = \int_{\mathbb{R}^d} \langle T_{m_s} f(x), g(x) \rangle dx,$$

where

$$m_s(\xi) = (1 - e^{-2|s|\Lambda(\xi)}) \frac{\int_{\mathbb{R}^d} (1 - \cos \xi \cdot z) \phi(z) \nu(dz) + \frac{1}{2} \int_{S^{d-1}} (\xi \cdot \theta)^2 \psi(\theta) \mu(d\theta)}{\Lambda(\xi)}$$

with the value 0 on $\{\Lambda = 0\}$. Indeed, each jump or Brownian contribution at time r contributes one factor $e^{-(0-r)\Lambda(\xi)}$ from the parabolic extension and another from the terminal test function; integrating over $r \in [s, 0]$ yields the factor $1 - e^{-2|s|\Lambda(\xi)}$ divided by the denominator. Hölder duality then gives

$$\|T_{m_s} f\|_{L^p(\mathbb{R}^d; X)} \leq C_{p,X} \|f\|_{L^p(\mathbb{R}^d; X)}.$$

Letting $s \rightarrow -\infty$, $m_s(\xi) \rightarrow m(\xi)$ pointwise. For $f \in C_c^\infty(\mathbb{R}^d; X)$, Plancherel on finite-dimensional ranges gives $T_{m_s} f \rightarrow T_m f$ in L^2 , and Fatou gives the same L^p bound for $T_m f$. Density extends T_m to all of $L^p(\mathbb{R}^d; X)$.

For a general Levy measure V , set $V_n = 1_{\{1/n \leq |z| \leq n\}} V$ and apply the finite-measure result to V_n . The integrability condition $\int (|z|^2 \wedge 1) V(dz) < \infty$ and dominated convergence imply pointwise convergence of the corresponding symbols. If V or μ is not symmetric, replace it by its symmetrization and replace the coefficient by the associated Radon–Nikodym average; since $1 - \cos(\xi \cdot z)$ and $(\xi \cdot \theta)^2$ are even, this leaves the symbol unchanged and preserves the bound $|\phi|, |\psi| \leq 1$. This proves the theorem.

For complex X and complex coefficients, apply the argument to the realification of X . Multiplication by a scalar of modulus at most one is a real-linear contraction, so the weak-subordination check above is unchanged. $\square$

Verification Notes and Limitations

- The proof uses only the general weak differential subordination theorem from [1] and the standard Bañuelos–Bogdan parabolic multiplier computation.
- The result is full for the selected concrete source route: it verifies the combination mechanism for the canonical mixed Levy/Gaussian symbol class. It does not claim a new class of multipliers beyond the earlier Theorem 4.1 of [2].

- The constant is deliberately not sharp. The direct general-WDS route inherits $C_{p,X} = \beta_{p,X}^2(\beta_{p,X} + 1)$, whereas the special proof in [2] gives $\beta_{p,X}$.
- Novelty search: local indexes were searched for arXiv:1706.01731, the open sentence, weak differential subordination, Fourier multipliers, and the nearby arXiv ids 1703.07817 and 1805.03948. Web exact-phrase searches for the open sentence and for a combined WDS multiplier proof did not find a later explicit answer.

Human Review Recommendation

Review the multiplier-symbol computation in the mixed parabolic pairing, especially the factor $1 - e^{-2|s|\Lambda(\xi)}$ and the symmetrization step for non-symmetric measures. The weak differential subordination step itself is straightforward and is the intended new mechanism of the packet.

References

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- [2] I. S. Yaroslavl'tsev. Fourier multipliers and weak differential subordination of martingales in UMD Banach spaces. arXiv:1703.07817, 2017.
- [3] R. Bañuelos and K. Bogdan. Levy processes and Fourier multipliers. *J. Funct. Anal.* 250(1):197–213, 2007.
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